

Better connections!

FESTO



Quick selection with our poster for tubing, fittings and flow control valves

Everything from a single source

Festo offers a precisely tailored and economical solution to all connection problems – plus reliability! Thanks to a wide variety of different resistance properties and characteristics we can meet virtually any requirement.

From flame-retardant to antistatic, from food industry compatibility to welding spatter resistance. You therefore only pay for the functions you need. Always included is the tried and tested Quick Star releasing principle based on Festo plug and work® – for quick, simple and reliable assembly.

For even greater safety:

Free of charge tubing selection software, media resistance information and Online Shop

Online Shop

- Quick and simple
- Find, compare and configure products
- Check prices and availability and order
- Transfer product from the catalogue into the shopping basket
- Import or copy and insert product lists
- Import part lists
- Save and call up shopping baskets again
- Track orders and re-order

Media resistance information

Which tubing is resistant to benzene?

Can elastomers withstand contact with glycol?

Which high-grade steel is suitable for use in acetic acid?

These and numerous other questions regarding the media resistance of different materials are answered quickly and easily here.

Tubing selection software

Simply enter your application-specific requirements and wait for the system to guide you quickly and reliably to the appropriate solution.



Process reliability and reduction in costs for low value items

Low value items have a significant influence on procurement and stock administration costs. In general, they represent a small proportion of total material costs but a large proportion of the material to be managed. Festo's Logistics Optimisation Service helps reduce the costs of stock administration and the ordering process while increasing process reliability at the same

time. Festo supports your logistics concept by providing independent modules from the Logistics Optimisation Service.

Select the suitable solution for your processes from the Festo logistics products.

The benefit to you

- Reduced order costs thanks to optimised ordering processes
- Fewer quality defects due a lower error rate
- Avoid unnecessary financial outlay thanks to a perfect parts range



Module 1: Analysis of the product range and the logistical process

- What products are ordered, how often and in what quantities?
- Simplifying or expanding the product range
- Finding potential for optimisation
- Planning of optimal storage and ordering quantities, as well as order cycles
- Annual statistics of your ordering behaviour

→ For details, see PSI 912.13



Module 2: Optimising product identification

- Stock labels with product illustrations make identification easier
- Coding user-specific information in all popular bar code and national languages facilitates integration into the customer's existing system

→ For details, see PSI 912.11



Module 3: Optimisation of the ordering process

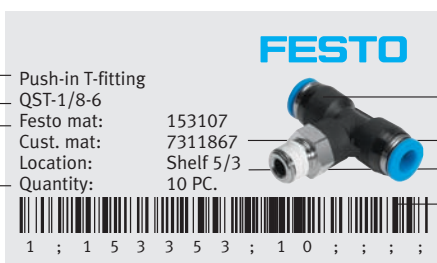
- Compiling pre-defined ordering quantities using bar code scanners
- Simple importing of scanned data into the Festo Online Shop
- 1-step ordering process – no waiting between request, price information and order

→ For details, see PSI 912.11

Festo stock labels – reliable, quick, individual

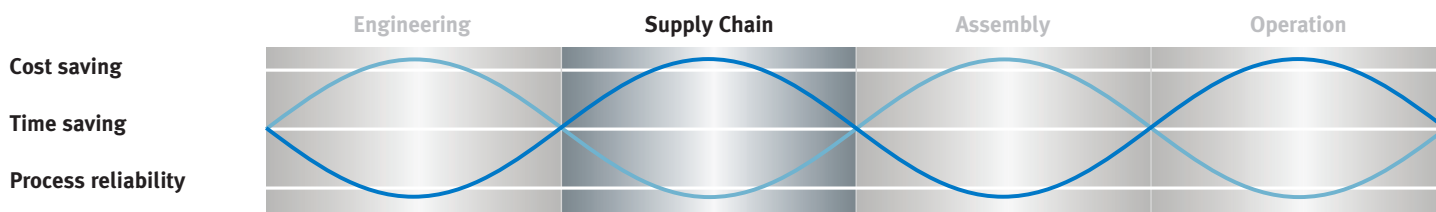
Standard specifications

Festo product designation — Push-in T-fitting
Festo type code — QST-1/8-6
Festo part number — Festo mat: 153107
Optimised ordering quantity — Cust. mat: 7311867
Location: Shelf 5/3
Quantity: 10 PC.



Optional specifications

Image or outline drawing
Customer material number
Location
Bar code



Cleverly combined:

Tubing-fitting combinations for all applications!

The right choice of the correct tubing-fitting combination is largely based on knowing its precise place of installation. To gain a quick understanding of the application, this can often be quickly determined by a single characteristic. One glance at the overview of our most popular tubing-fitting combinations enables you to make a quick and targeted choice.

Tubing

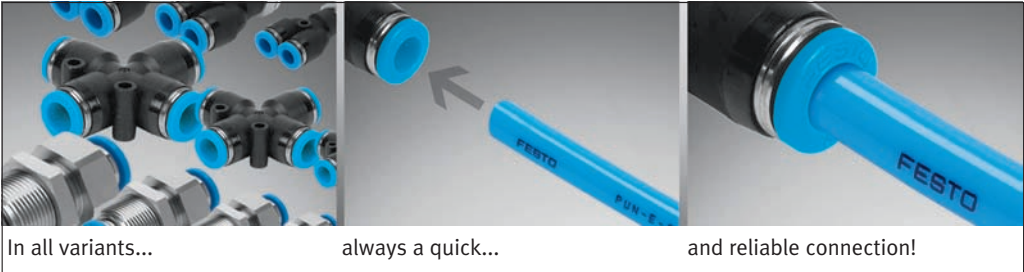
	Polyurethan (PU)						Polyethylen (PE)		Polyamid (PA)		PFA**	
Resistance and suitability	PUN/PUN-DUO	PUN-H/ PUN-H DUO	PUN-E	PUN-CM	PUN-V0	PUN-V0-C	PLN	PEN	PAN	PAN-R	PAN-V0	PFAN
Media	Compressed air, vacuum	Compressed air, vacuum water*	Compressed air, vacuum	Compressed air, vacuum	Compressed air, vacuum, water*	Compressed air, vacuum, water*	Compressed air, vacuum, water*, cleaning	Compressed air, vacuum	Compressed air, vacuum	Compressed air, vacuum	Compressed air, vacuum water*	Compressed air, vacuum water*, acid, alkalis agents
Temperature [°C]	-35...60	-35...60	-35...50	-35...60	-35...60	-35...60	-30...80	-30...60	-30...80	-30...80	-30...90	-20...150
Operating pressure, temperature dependent [bar]	-0.95...10	-0.95...10	-0.95...8	-0.95...10	-0.95...10	-0.95...10	-0.95...14	-0.95...10	-0.95...19	-0.95...35	-0.95...12	-0.95...16
Chemical resistant	+	+	+	+	+	+	++	++	++	++	-	+++
Hydrolysis resistant	+	+++	+++	+	+	+	+++	+++	++	++	+	+++
Antistatic	-	-	-	+++	-	-	-	-	-	-	-	-
Flame-retardant UL94 V0-V2	-	-	-	-	+++	+++	-	-	-	-	+++	+++
Food industry approval	-	+++	-	-	-	-	+++	-	-	-	-	+++
Halogen-free	+++	+++	+++	+++	+++	+++	+++	+++	-	-	-	-

** Perfluoralkoxy

Fitting

	Plastic/brass			All-metal	Stainless steel	Metal	Plastic
Resistance and suitability	QS(M)	QS(M)-B	QS-V0	QS-F	CRQS	NPQM	NPQP
Media	Compressed air, vacuum, water*	Compressed air, vacuum	Compressed air, vacuum, water*	Compressed air, vacuum, water*	Compressed air, vacuum, water* acid, alkalis	Compressed air, vacuum	Compressed air, vacuum
Temperature [°C]	-10...80	-10...60	0...60	0...150	-15...120	-20...70	-20...60
Operating pressure, temperature dependent [bar]	-0.95...14	-0.95...10	-0.95...10	-0.95...16	-0.95...10	-0.95...16	-0.95...10
Chemical resistant	+	+	+	++	+++	+	+++
Hydrolysis resistant	+	+	+	+++	+++	++	+++
Antistatic	-	-	-	+++	+++	+++	-
Flame-retardant UL94 V0-V2	-	-	+++	+++	+++	++	-
Food industry approval	-	-	-	++	+++	-	-
Halogen-free	+++	+++	+++	+++	+++	+++	+++

+++ very suitable ++ quite suitable + suitable - not suitable * water and other media on request



	Application	Tubing	Fitting	Details
	Economical I	PEN	QS-B GRLA-QS-B-20	The universally suitable combination at an attractive price. Can be widely used thanks to highly resistant materials. Easy to install thanks to optimised bending radii.
	Economical II	PEN	NPQM VFOC	The universal solution for metal fittings. Perfect for standard pneumatics applications in many different fields.
	Wide choice of variants	PUN	QS GRLA-QS-RS-D	More than 1000 types for maximum flexibility in standard applications.
	Hydrolysis resistant	PUN-H	QS GRLA-QS-RS-B	For applications in humid environments or in contact with water at up to 60 °C. PUN-H – microbe resistant thanks to modified material.
	Pressure resistant I	PAN	QS GRLA-QS-RS-B	Safe connection when used in pressure ranges of up to 14 bar. Economical in pneumatic installations operating in the high pressure range.
	Pressure resistant II	PAN-R	QS-F GRLA-F-QS-D	The high performance combination for applications in pressure ranges of up to 16 bar: For example, applications using the pressure booster DPA.
	Heat resistant	PFAN	QS-F GRLA-F-QS-D	For a reliable compressed air supply in high-temperature areas. Up to 10 bar at 80 °C or 6 bar at 150 °C – maximum process security at all times.
	Antistatic	PUN-CM	NPQM GRLA-F-QS-D	Electrically conductive tubing combined with an all-metal fitting.
	Food industry approved I	PUN-H	QS-F CRGRLA	FDA approval for use in the food industry – a hydrolysis resistant combination with enhanced functionality.
	Food industry approved II	PFAN	CRQS CRGRLA	Created for the highest standards, thanks to the highest corrosion resistance: this combination excels wherever maximum food hygiene is needed.
	Resistant to cleaning agents	PLN	NPQP	The cost-effective alternative to stainless steel is highly resistant to all usual cleaning agents and has maximum corrosion protection. That makes it perfect for critical environments such as splash zones.
	Media resistant	PFAN	NPQP	Completely resistant to all cleaning agents and lubricants and even permits the conveyance of acids and lyes without any problems.
	Flame retardant	PUN-VO	NPQM	Safe in areas where there is a risk of fire thanks to flame retardant properties to UL94.
	Resistant to welding spatter I	PUN-VO-C	QS-F GRLA-F-QS-D	The economical combination for applications in indirect proximity to welding applications. Safe thanks to a tubing wall thickness of 2 mm for all diameters.
	Resistant to welding spatter II	PAN-VO	QS-VO	Safe even in indirect proximity to welding spatter. Two-layer tubing and special fitting.

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This page provides an overview of all included posters!

QS	Poster	Types	Applications
	Standard fittings Part 1 von 2	• Mini types • Standard types • Straight and L- shaped • Push-in bulkhead fittings • Push-in fittings, self-sealing	• Wide range of variants • Hydrolysis resistant • Pressure resistant
QS	Poster	Types	Applications
	Standard fittings Part 2 von 2	• L, T, Y-shaped • Multiple distributor	• Wide range of variants • Hydrolysis resistant • Pressure resistant
QS	Poster	Types	Applications
	Standard connectors	• Straight, T, L, X, Y-shaped • With push-in sleeve • Push-in bulkhead fittings and push-in fittings, self-sealing	• Wide range of variants • Hydrolysis resistant • Pressure resistant • Economical
QS-B	Poster	Types	Applications
	Fittings connectors for core functions	• Mini types • Straight, T, L, X, Y-shaped	• Economical
NPQM	Poster	Types	Applications
	Metallic push-in fittings	• Push-in fittings with internal hex, thread, T, L-shaped • Plug connector straight, with sleeve, L, T, Y-shaped	• Economical • Antistatic • Flame-retardant
NPQP	Poster	Types	Applications
	Polypropylene fittings	• Plug connectors, L, T, Y-shaped • Plug connectors, L, T, Y-shaped • Bulkhead connector • Blanking plugs	• Cleaning agent • Media resistant
QS-F CRQS QS-V0	Poster	Types	Applications
	Metal and special fittings/connectors	• Metal, stainless steel, • Straight, L and T-shaped	• Pressure/heat resistant • Antistatic, Food industry approved • Cleaning agent/media resistant • Flame-retardant • Welding spatter resistant
PUN-... PAN-... PEN PFAN PLN	Poster	Types	Applications
	Plastic tubing	• Standard O.D. • Different materials • Standard and bulk packs	• For all applications
NPFB-... GRLA-.... GRLZ-... GRLO-... VFOC/VFOV-...	Poster	Types	Applications
	Threaded fittings and flow control valves	• Threaded fittings, • Nickel-plated brass • Function fittings	• For all applications

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Technical data for flow control valves

Designation		GRLA-(RS)-B	GRLA-PK-(RS)-B	GRLA-QS-RS-B	GRLA-QS-(RS)-D	VFOC	VFOV	GRLA-F-QS-D	CRGRLA
Function	Exhaust air flow control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Supply air flow control	Yes	Yes	-	Yes	-	Yes	-	-
	Without non-return function	-	-	-	-	-	-	-	-
Actuation	With tool	Yes	Yes	Yes	Yes	Yes	-	Yes	Yes
	Without tools	Yes	Yes	Yes	Yes	-	Yes	-	-
Media		Compressed air, Vacuum	Compressed air, Vacuum	Compressed air, Vacuum	Compressed air, Vacuum	Compressed air, Vacuum	Compressed air, Vacuum	Compressed air, Vacuum	Compressed air, Vacuum
Operating pressure [bar]		0.3 ... 10	0.3 ... 11	0.2 ... 10	0.2 ... 10	0.2 ... 10	0.2 ... 10	0.2 ... 10	0.3 ... 10
Operating temperature [°C]		-10 ... +60	-10 ... +60	-10 ... +60	-10 ... +60	-10 ... +60	-10 ... +60	0 ... +150	-20 ... 80
Mounting position		Any	Any	Any	Any	Any	Any	Any	Any
Hydrolysis resistant		++	++	+++	++	++	+++	+++	+++
Antistatic		++	++	-	++	++	-	+++	+++
Flame-retardant		++	++	-	-	-	-	+++	+++
Food industry		-	-	-	-	-	-	++	++

+++ very suitable ++ quite suitable + suitable – not suitable